nature portfolio

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Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☒	☐	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐	A description of all covariates tested
☒	☐	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☒	☐	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☒	☐	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No software was used, data was retrieved from the ProteomeXchange consortium (http://proteomecentral.proteomexchange.org) via the PRIDE partner repository with the dataset identifier PXD061173
Data analysis	pLink 3.0.17, Scout 2.0.0 beta, XlinkX in Proteome Discoverer 3.2.0.450 was used to identify crosslinks; pyXLMS (proposed in this paper) was used for data processing, available from https://github.com/hgb-bin-proteomics/pyXLMS; xiVIEW, AlphaLink2 version git@d54bcbf, XMAS version 1.1.3 in ChimeraX version 1.10.1 were used for down-stream analysis with pyXLMS outputs;

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

No data was measured as part of this study. Data was retrieved from a public data repository, namely rom the ProteomeXchange consortium (http://proteomecentral.proteomexchange.org) via the PRIDE partner repository with the dataset identifier PXD061173.

The with pyXLMS processed data files and result files are available via https://github.com/hgb-bin-proteomics/pyXLMS-manuscript

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	not applicable, no human or animal data has been used
Reporting on race, ethnicity, or other socially relevant groupings	not applicable, no human data has been used
Population characteristics	not applicable
Recruitment	not applicable
Ethics oversight	not applicable

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No sample size calculation was performed, data was retrieved from an already published dataset, see https://doi.org/10.1038/s42004-025-01644-6
Data exclusions	Only a subset of the dataset was used as the aim of this publication is to showcase the technical capabilities of the software. For that we chose the Cas9 crosslinked sample enriched with Cytiva beads. No data was excluded from the chosen sample.
Replication	All experiments were performed using 3 replicates of which all were successful, see https://doi.org/10.1038/s42004-025-01644-6
Randomization	Not applicable, as we do not investigate biological effects and therefore had no groups. Replicate MS measurements of benchmark studies where measured in alternating order to avoid effects originating from altered MS performance with time. See https://doi.org/10.1038/s42004-025-01644-6
Blinding	As this is not a clinical study, blinding is not applicable.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	not applicable
Novel plant genotypes	not applicable
Authentication	not applicable